AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

AKU-EB / EXAMINATIONS 2016

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

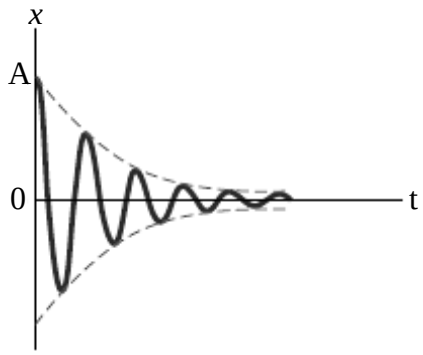
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

A graph of displacement versus time for an oscillatory system is shown below.



a. State the type of oscillatory system shown in the graph. (1 Mark)

_____

_____

b. What does the graph suggest? (1 Mark)

_____

_____

c. Write the reason of the phenomenon shown in the graph by mentioning TWO points. (2 Marks)

_____

_____

_____

_____

AKU-EB MAY EXAMINATIONS 2016

Q.2. (Total 2 Marks)

State laws of refraction.

_____

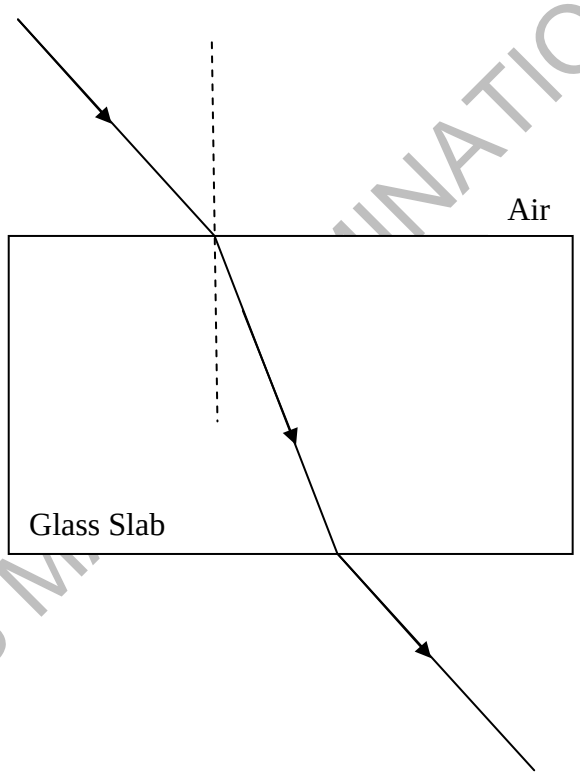
_____

_____

_____

Q.3. (Total 3 Marks)

Label the given diagram by putting (normal, angle of incidence and refracted ray) in appropriate positions.



AKU-EB M... EXAMINATIONS 2016

Q.4. (Total 3 Marks)

Two different capacitors of capacitance 6 μF and 12 μF are connected in series combination. If a battery of 12 V is connected with the circuit, then find the equivalent capacitance of the combination. Also draw a circuit diagram of this combination.

Space for Circuit Diagram

Q.5. (Total 3 Marks)

According to the safety measures, arrange the given points so that a fuse can protect an electrical circuit.

- fuse wire heats up
- circuit is broken so current stops
- current becomes too high

1. _____

2. _____

3. _____

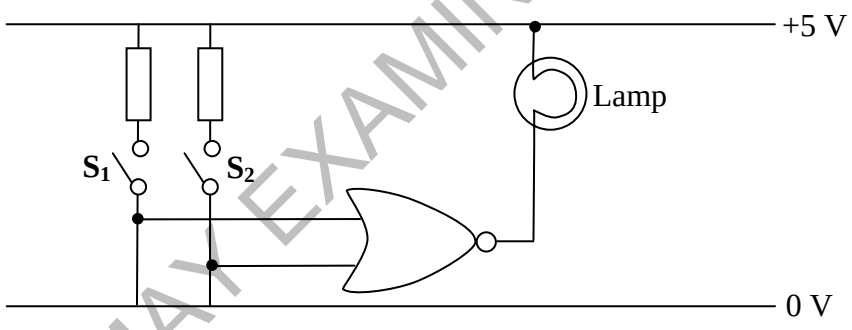
AKU-EB MAY EXAMINATIONS 2016

Q.6. (Total 3 Marks)

State TWO conditions that must be fulfilled for a charge to experience magnetic force when it is placed in a magnetic field and state how the magnetic force could be varied.

Q.7. (Total 3 Marks)

The given diagram shows a logic gate connected in a circuit.



a. Make a truth table for the above logic gate. (2 Marks)

S_1	S_2	Output

b. How would the switches S_1 and S_2 light the lamp? (1 Mark)

AKU-EB MAY EXAMINATIONS 2016

Q.8. (Total 4 Marks)

If the half-life of Krypton is 3.16 minutes, then how many grams of Krypton out of 100 grams will remain disintegrated after 9.48 minutes.

AKU-EB MAY EXAMINATIONS 2016

Q.9. (Total 5 Marks)

EITHER

a. How are mechanical and electromagnetic waves different from each other? Describe in FIVE points.

OR

b. Acoustics is the study of waves, vibrations and sound. Acoustic protection is the application of soft and porous material to protect individuals against undesirable sounds and noises.

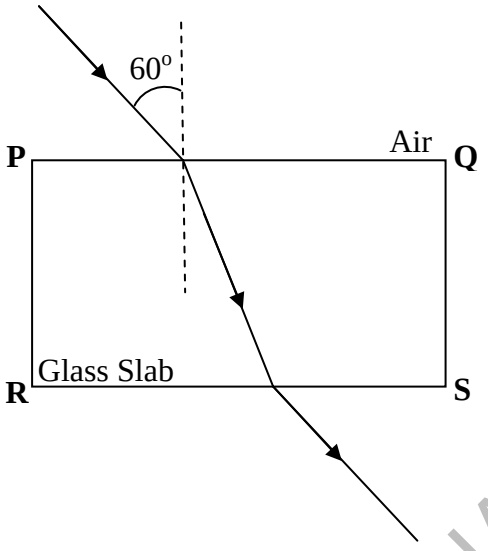
Describe any FIVE points to highlight the significance of acoustic protection.

AKU-EB MAY EXAMINATIONS 2016

Q.10. (Total 5 Marks)

EITHER

a. The given diagram shows a glass slab **PQRS**. A ray of light incident on the side **PQ** of the glass slab is making an angle of 60°.



If the refractive index of the glass slab is 1.5 and value of $\sin 60^o$ is 0.866, then calculate the angle of refraction and sine of critical angle ($\sin\angle C$) for the given glass slab.
(Where $\sin^{-1}(0.577) = 35.2^o$)

OR

b. An object is placed at a distance of 12 cm from a convex lens of focal length 8 cm. Find the position and state the nature of image.

(**Note:** Extract data from the given question and provide your answer in SI unit.)

AKU-EB MAY EXAMINATIONS 2016

AKU-EB MAY EXAMINATIONS 2016

Q.11. (Total 5 Marks)

EITHER

a. Describe in any FIVE points, how electrostatic induction is used in spray painting and in the extracting of dust from dirty carpets and from the chimneys of industrial machinery.

OR

b. Describe any FIVE safety measures that should be taken in connection with the household circuit.

AKU-EB MAY EXAMINATIONS 2016

END OF PAPER

Please use this page for rough work

AKU-EB MAY EXAMINATIONS 2016

Please use this page for rough work

AKU-EB MAY EXAMINATIONS 2016